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CASUAL FEATURE ATTRIBUTION FOR HIGH-STAKES TABULAR DECISIONS WITH ACTIONABLE RECOURSE

A Machine Learning and Explainable AI Based AI

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PROJECT GUIDE :
DESIGNATION:

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ABSTRACT

- Machine Learning models are widely used in high-stakes applications such as healthcare and finance.
- Many existing models lack transparency and fail to explain the reasons behind their predictions.
- Traditional explainability techniques like SHAP and LIME identify influential features but do not establish causal relationships.
- The proposed framework combines Explainable AI (XAI), Causal Inference and Actionable Recourse.
- It aims to identify influential factors, investigate causal relationships and generate realistic recommendations.
- The ultimate goal is to make Machine Learning decisions more transparent, interpretable and actionable.

Aim & Objective

AIM : To develop an Explainable AI framework using Causal Inference and Actionable Recourse to interpret Machine Learning predictions , identify underlying factors and generate realistic recommendations for improving decision making

- **Data Collection & Preprocessing:** To collect, clean and prepare relevant tabular datasets for Machine Learning.
- **Model Development:** To develop and train Machine Learning models for high-stakes decision prediction.
- **Causal Analysis & Explainability:** To identify influential features and investigate causal relationships using XAI and causal inference techniques.
- **Actionable Recourse & Evaluation:** To generate feasible counterfactual recommendations and evaluate their effectiveness and reliability.

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Introduction & Problem Statement

Background & Context

AI-based systems are increasingly used in high-stakes decisions such as loan approvals and credit risk assessment. However, complex ML models often lack transparency. Causal inference and explainable AI offer opportunities to make these decisions more understandable and actionable.

Problem Statement

Existing explanation methods identify features influencing predictions but often fail to distinguish correlation from causation or suggest realistic actions to change an unfavourable decision.

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Introduction & Problem Statement

Dynamic Aspect	Existing State / Challenge	Consequence
Decision Transparency	Complex ML models provide limited decision explanations.	Lack of trust and interpretability.
Causal Validity	SHAP and LIME rely primarily on feature associations.	Explanations may not reflect true causal effects.
Actionability	Counterfactual methods may suggest unrealistic changes.	Impractical recommendations for applicants.
Feature Constraints	Mutable and immutable features are not always distinguished.	Suggestions may involve unchangeable attributes.

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Literature Survey

Ref No.	Author & Year	Paper / System Title	Methodology	Key Limitation
[1]	Lundberg & Lee (2017)	*A Unified Approach to Interpreting Model Predictions (SHAP)*	SHapley Additive Explanations	Correlational, not causal; misleading under confounders
[2]	Ribeiro et al. (2016)	*"Why Should I Trust You?": Explaining the Predictions of Any Classifier (LIME)*	Local linear surrogate approximation	Unstable; does not reflect causal mechanisms
[3]	Ustun et al. (2019)	*Actionable Recourse in Linear Classification*	LP-based algorithmic recourse	Limited to linear models; ignores causal constraints

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Literature Survey

Ref No.	Author & Year	Paper / System Title	Methodology	Key Limitation
[4]	Wachter et al. (2017)	*Counterfactual Explanations Without Opening the Black Box*	Distance minimisation (nearest counterfactual)	No causal feasibility enforcement; unrealistic suggestions
[5]	Karimi et al. (2021)	*Algorithmic Recourse: From Counterfactual Explanations to Interventions*	SCMs + causal intervention framework	No deployable UI; computationally expensive
[6]	Mothilal et al. (2020)	*Explaining Classifiers through Diverse Counterfactual Explanations (DiCE)*	Diversity-constrained counterfactual generation	Violates real-world causal constraints (e.g., age)

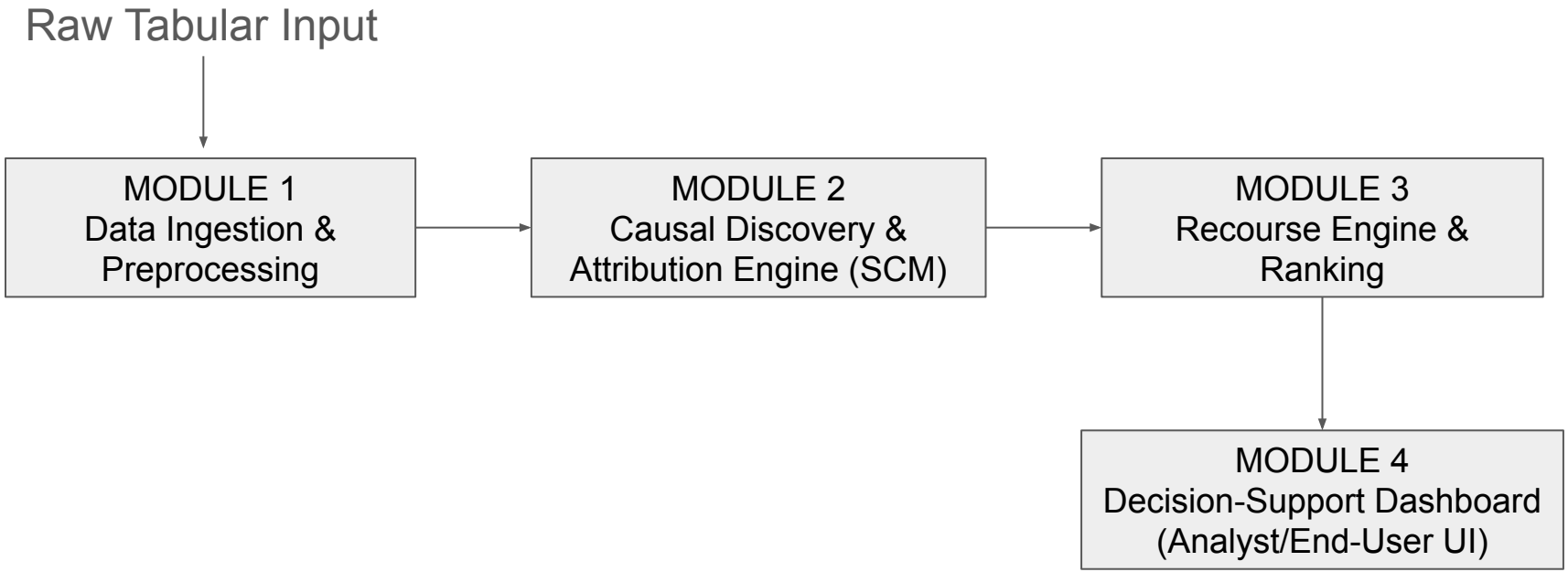
Limitations of Existing Systems

1. **Correlational, not causal attribution**: SHAP/LIME measure statistical association, not intervention effect; attributions are unreliable under feature dependencies and confounders.
2. **Causally infeasible recourse**: Counterfactual methods minimise feature-space distance without respecting the SCM, producing suggestions that are mathematically valid but practically impossible.
3. **No integrated decision-support interface**: Research prototypes lack production-ready dashboards for loan officers, clinicians, or compliance analysts to act on recourse in their workflow.
4. **Scalability constraints**: SCM-based causal recourse (Karimi et al.) is prohibitively costly for real-time inference over high-dimensional enterprise datasets.
5. **No actionability filtering**: None of the reviewed systems systematically distinguish mutable from immutable features, reducing practical utility.

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Proposed System Architecture

System Architecture Diagram:



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Proposed System Architecture

Core Methodology & Algorithmic Process:

1. **Data Ingestion & Preprocessing**: Tabular records (loan/patient/employee data) are ingested, type-validated, imputed, normalised, and annotated with mutable/immutable feature flags per a domain configuration schema.
2. **Causal Graph Construction**: A Directed Acyclic Graph (DAG) is learned using the **PC Algorithm** (constraint-based) or **NOTEARS** (gradient-based). Expert-specified domain edges are incorporated as hard constraints, producing a full Structural Causal Model (SCM).
3. **Causal Feature Attribution**: Average Causal Effect (ACE) and Individual Causal Effect (ICE) are computed per feature using $do(X_i = x_i)$ (Pearl's do-calculus), quantifying the true interventional impact on the prediction, not correlation.
4. **Actionable Recourse Generation**: A constrained optimisation problem identifies the minimum-cost set of interventions on mutable features that flips an adverse decision; candidates are filtered for causal feasibility, actionability, and distributional plausibility, then ranked by implementation cost.
5. **Payload Delivery & Dashboard**: All attributions, causal path visualisations, and ranked recourse steps are rendered in a **Streamlit/React decision-support UI** with audit logging for regulatory compliance.

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Proposed System Architecture

Component	Technology	Role
Causal Discovery	PC Algorithm / NOTEARS	Learns the causal DAG from data
Core Classifier	XGBoost / LightGBM	High-accuracy base prediction model
Causal Attribution	SCMs + do-calculus	Computes true causal feature effects
Recourse Generator	Constrained causal optimisation	Produces feasible, actionable recourse paths

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Proposed System Architecture

Component	Technology	Role
Actionability Filter	Domain schema + mutability flags	Excludes immutable features from recourse
Backend API	FastAPI (Python)	Serves inference & attribution endpoints
UI Dashboard	Streamlit / React.js	Analyst-facing interactive interface
Audit & Logging	PostgreSQL	Regulatory compliance audit trail

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Implementation Tech Stack & Tools

Category	Tools / Technologies Selected	Justification / Role
Programming Language	Python 3.10+	Core language for ML model development, causal inference, optimization and backend implementation.
Data processing	Numpy, Pandas	Numerical computation, dataset manipulation, preprocessing and feature preparation.
Machine Learning	Scikit Learn	Implements classification algorithms such as Random Forest and Logistic Regression for decision prediction.
Causal Inference Libraries	DoWhy, causal-learn	Supports Structural Causal Model construction, causal effect estimation and intervention analysis.

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Implementation Tech Stack & Tools

Resource Optimization	Scipy	Solves constrained optimization problems to generate minimal, feasible feature modifications.
Explainability	SHAP, Matplotlib	Visualizes feature contributions and compares conventional explanations with causal attribution.
Backend API	FastAPI	Exposes prediction, causal attribution and recourse generation modules through RESTful endpoints.
Frontend Dashboard	Streamlit	Provides an interactive decision-support interface for visualizing predictions, causal explanations and recourse suggestions.

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Implementation Tech Stack & Tools

Database & Audit Logging	PostgreSQL	Stores applicant records, prediction results, generated recommendations and audit trails.